

Achuta Kadambi

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Bio: Achuta Kadambi (PhD, MIT '18) is a tenured Associate Professor at UCLA in Electrical Engineering and Computer Science. He leads an AI research group that focuses on computer vision and spatial intelligence. He is the recipient of early career awards including from NSF (CAREER), DARPA (YFA), ARO (YIP), IEEE (HKN under 35 award), and Forbes (30 under 30). He has co-founded two California companies: Akasha Imaging (acquired in 2022, now with Alphabet) and Vayu Robotics (acquired in 2025, now with Nasdaq: SERV). Kadambi is an inventor on 40+ issued US patents, and a co-author of the textbook Computational Imaging in MIT Press.

Education

PhD	MIT ¹	2018
MS	Yale	2012
BS	Berkeley	2011

¹ Interdepartmental Doctoral Degree Program between the MIT Media Lab and MIT EECS

Academic Appointments

Associate Professor	UCLA	2024-
Assistant Professor	UCLA	2018-2024

Start-up Companies Founded

Co-founder	Vayu Robotics (acquired '25, now w/ Nasdaq: SERV)	2022-2025
Co-founder	Akasha Imaging (acquired '22, now / Alphabet)	2018-2022

Awards

2024	Senior Member, IEEE	UCLA
2022	IEEE-HKN Outstanding Young Professional under 35 Award	UCLA
2022	Best Demo Award, Honorable Mention, UIST	UCLA
2021	NSF CAREER Award	UCLA
2021	DARPA Young Faculty Award	UCLA
2021	Army Research Office Young Investigator Award (ARO YIP)	UCLA
2021	National Academy of Engineering (NAE) Frontiers of Engineering	UCLA
2020	Google Faculty Award	UCLA
2020	Senior Member National Academy of Inventors	UCLA
2019	Forbes 30 under 30, Science	UCLA
2019	NSF CRII Research Initiation Award	UCLA
2019	Sony Imaging Young Faculty Award	MIT
2018	Best Paper Award, ICCP	MIT
2016	Lemelson-MIT Student Prize	MIT
2016	Rahamimoff Award, US-Israel Science Foundation	MIT
2016	Best Papers Special Issue Selection, ICCV	MIT
2016	Best Presentation Award, CVPR VIEW	MIT
2015	World Changing Idea, Scientific American	MIT
2014	Qualcomm Innovation Fellowship	MIT
2013	Draper Fellowship	MIT

Awards won by UCLA Students

2025	PhD Dissertation Year Award, UCLA (Shijie Zhou)
2025	Amazon-UCLA PhD Fellow (Rishi Upadhyay)
2023	Distinguished Master's Thesis Award, ECE Department (Adnan Armouti)
2022	NSF Graduate Research Fellowship GRFP, Winner (Ellin Zhao)
2022	NSF Graduate Research Fellowship GRFP, Hon. Mention (Sasha Vilesov)
2021	Cisco PhD Fellowship (Pradyumna Chari)
2020	Guru Krupa Graduate Fellowship, UCLA (Chinmay Talegaonkar)
2019	Best Undergrad Demo, Annual Research Review (A Padhye, A Tilaye, et al)
2019	Best Poster, runner up, SoCal Machine Learning Day (Yunhao Ba)

Invited Talks

2025 UNAM Image Processing Group Group (Mexico City, MX)
2025 Army Futures Command, Army Research Lab (Delphi, MD)
2024 **Keynote**, Human-centered AI Conference (Malibu CA)
2024 Center for AI, Universidad Torcuato Di Tella (Buenos Aires, Argentina)
2024 MIT Media Lab (Cambridge MA)
2024 Invited Speaker, DARPA Triage Challenge (DTC) Meeting
2024 **Keynote**, CVPR 2024, UG2 Workshop (Seattle WA)
2024 Invited Speaker, Army Research Lab (Playa Vista CA)
2023 Lecturer, SIGGRAPH 2023, Course on Polarization (Los Angeles CA)
2023 Lecturer, CVPR 2023, Tutorial on Polarization (Vancouver Canada)
2023 Invited Speaker, SID Display Conference (Los Angeles CA)
2023 UC Davis, Bioengineering Seminar (Davis, CA)
2022 **Keynote**, CVPR UG2 Workshop (New Orleans, LA)
2022 **Keynote**, CVPR CVPM Workshop (New Orleans, LA)
2022 University of Maryland, Computer Science (Wash. DC)
2022 CMU Bioengineering Department Seminar (Pittsburgh, PA)
2022 Optical Sensors and Sensing Congress (Vancouver, CA)
2021 MIT AeroAstro Department (Cambridge, MA)
2021 Boston University Electrical Engineering Department (Boston, MA)
2021 EPFL, Lausanne Switzerland (Rescheduled - COVID)
2021 UCLA Medical School Grand Rounds, Los Angeles CA (Los Angeles, CA)
2021 Snap Inc. (Virtual)
2021 Cornell AI in Medicine Seminar (Virtual)
2021 ETH Zurich Computer Vision Seminar (Zurich, Switzerland)
2021 Amazon + UCLA Science Hub Kickoff Event (Los Angeles, CA)
2021 Pixel Cafe at UCSD (Virtual)
2021 UC Berkeley Bioengineering, Guest Lecturer for Medical Devices (Virtual)
2021 Black in Neuro Panel, Imperial College London (Virtual)
2021 Army Research Lab A2I2 Summit (Virtual)
2021 ICCV GigaVision Workshop (Virtual)
2021 CLEO Panel on AI and Photonics (Virtual)
2021 Army Research Lab Workshop on Synthetic Data (Virtual)
2020 SPIE Workshop on Computational Imaging (Virtual)
2020 Army Research Lab, Adelphi MD
2020 CVPR Visual Physics, Seattle WA
2019 DARPA/MEC workshop on AI, San Jose CA
2019 Stanford EE Department, Stanford CA
2019 MIT Media Lab, Cambridge MA
2019 Lemelson-MIT EurekaFest, Cambridge MA
2019 Computational Light Transport Summit, Banff Canada
2019 Indian Institute of Science, EE Department, Bangalore India
2019 Machine Learning Summer School, Bangalore India
2019 Honeywell Technology Symposium, Phoenix AZ
2019 Annual Research Review, UCLA, Los Angeles CA
2018 University of California, Los Angeles CA
2018 Carnegie Mellon University, Pittsburgh PA

2018 MIT CSAIL, Cambridge MA
2017 University of Tokyo, Tokyo JP
2017 Cymer Semiconductor Equipment, San Diego CA
2017 Computer Vision and Information Processing Society of Japan, Nagoya JP
2016 Honeywell Technology Symposium, Phoenix AZ
2016 Columbia CS, New York City, NY
2016 Cornell Tech, CS New York City, NY
2016 Mitsubishi Electric Research Lab (MERL), Boston MA
2016 University of Pennsylvania GRASP Lab, Philadelphia PA
2016 Princeton CS, Princeton, New Jersey
2016 Weizmann Institute of Science, Rehovot Israel
2016 Technion CS Department, Haifa Israel
2016 Mass General Hospital (MGH), Boston MA
2016 OSA Invited Talk, Heidelberg Germany
2016 Analog Devices, Cambridge MA
2015 Computational Imaging Summit, Dagstuhl Germany
2015 Microsoft Research, Redmond WA
2014 Qualcomm Research, San Diego CA
2014 Technion, Haifa Israel
2014 Microsoft iToF Workshop, Ein Gadi Israel
2014 IIT-Bombay, Bombay India
2013 Nokia Research, Bangalore India

Professional Service

Co-organizer, ICCV '25 End-to-End 3D Learning Workshop
NSF CISE Panelist, 2025, 2023, 2022
COSMOS-UCLA High School Summer Camp Co-Instructor, 2025, 2024
UCLA Ad Hoc Committee for Artificial Intelligence in Teaching and Learning, 2024
UCLA ECE FastTrack Adviser, 2025, 2024
Co-instructor, "Polarization-based Visual Computing", SIGGRAPH 2023
Co-instructor, "Polarization-Based Computer Vision", CVPR 2023
UCLA ECE TA Awards Committee 2023
UCLA, ECE PhD thesis award committee 2023
Industry Chair, ICCP 2023, ICCP 2022, ICCP 2020
Program chair, CVPR CCD 2021, 2020
Program committee, Pacific Graphics 2019
Program committee, CVPR 2021, 2020, 2019, 2018
Program committee, ICCP 2023, 2022, 2021, 2020, 2019, 2018
Program committee, ICCV PBDL Workshop 2017
Reviewer, SIGGRAPH, SIGGRAPH Asia, ICCV, CVPR, ECCV, ICCP, etc.
MIT, undergrad admissions committee
Lemelson-MIT student prize selection committee

Textbook

- T1 A. Bhandari, A. Kadambi, R. Raskar, Computational Imaging (450 pages). MIT Press, 2022 (E-PDF at imagingtext.github.io)

Papers Published (CS conference or journals)

Note; In computer science (CS), top conferences like CVPR, ICCV, and ECCV are considered on par with journals in terms of prestige and peer review rigor. As such, the list below includes both journals and top CS conference papers.

<https://scholar.google.com/citations?user=UMzWJikAAAAJ&hl=en&oi=ao>

- P1 Zhou, S., Vilesov, A., He, X., Wan, Z., Zhang, S., Nagachandra, A., Chang, D., Chen, D., Wang, X. and Kadambi, A. VLM4D: Towards Spatiotemporal Awareness in Vision Language Models. ICCV 2025
- P2 Li, R., Pan, P., Yang, B., Xu, D., Zhou, S., Zhang, X., Li, Z., Kadambi, A., Wang, Z., Tu, Z. and Fan, Z. 4k4dgen: Panoramic 4d generation at 4k resolution. ICLR 2025
- P3 Zhang, H., Alaluf, Y., Ma, S., Kadambi, A., Wang, J. and Aberman, K. InstantRestore: Single-Step Personalized Face Restoration with Shared-Image Attention. SIGGRAPH 2025 (conference).
- P4 Chari, P., Ma, S., Ostashev, D., Kadambi, A., Krishnan, G., Wang, J. and Aberman, K. Personalized restoration via dual-pivot tuning. IEEE Transactions on Image Processing 2025.
- P5 Park, S., Vilesov, A., Zhang, J., Khalili, H., Tian, Y., Kadambi, A. and Sehatbakhsh, N. Chimera: Creating Digitally Signed Fake Photos by Fooling Image Recapture and Deepfake Detectors. USENIX Security 2025.
- P6 Zhou, S., Ren, H., Weng, Y., Zhang, S., Wang, Z., Xu, D., Fan, Z., You, S., Wang, Z., Guibas, L. and Kadambi, A. Feature4x: Bridging any monocular video to 4d agentic ai with versatile gaussian feature fields. CVPR 2025
- P7 Ezhov, V., Park, H., Zhang, Z., Upadhyay, R., Zhang, H., Chandrappa, C.C., Kadambi, A., Ba, Y., Dorsey, J. and Wong, A. All-day depth completion. IROS 2024.

- P8 Wang, Z., Xu, Q., Tan, F., Chai, M., Liu, S., Pandey, R., Fanello, S., Kadambi, A. and Zhang, Y. Mvdd: Multi-view depth diffusion models. ECCV 2024
- P9 Chari, P., Harish, A.B., Armouti, A., Vilesov, A., Sarda, S., Jalilian, L. and Kadambi, A., Implicit neural models to extract heart rate from video. ECCV 2024
- P10 Zhou, S., Fan, Z., Xu, D., Chang, H., Chari, P., Bharadwaj, T., You, S., Wang, Z. and Kadambi, A. Dreamscene360: Unconstrained text-to-3d scene generation with panoramic gaussian splatting. ECCV 2024
- P11 Fan, Z., Zhang, J., Cong, W., Wang, P., Li, R., Wen, K., Zhou, S., Kadambi, A., Wang, Z., Xu, D., Ivanovic, B. and Wang, Y. Large spatial model: End-to-end unposed images to semantic 3d. NeurIPS 2024
- P12 Zhou, S., Chang, H., Jiang, S., Fan, Z., Zhu, Z., Xu, D., Chari, P., You, S., Wang, Z. and Kadambi, A. Feature 3dgs: Supercharging 3d gaussian splatting to enable distilled feature fields. CVPR 2024 (Highlight Paper)
- P13 Upadhyay, R., Zhang, H., Ba, Y., Yang, E., Gella, B., Jiang, S., Wong, A. and Kadambi, A. Enhancing Diffusion Models with 3D Perspective Geometry Constraints. ACM SIGGRAPH Asia 2023 (journal)
- P14 Wang, Z., Zhou, S., Park, J., Paschalidou, D., You, S., Wetzstein, G., Guibas, L. and Kadambi, A. ALTO: Alternating Latent Topologies for Implicit 3D Reconstruction. CVPR 2023
- P15 Zhang, H., Ba, Y., Yang, E., Mehra, V., Gella, B., Suzuki, A., Pfahnl, A., Chandrappa, C., Wong, A. and Kadambi, A. WeatherStream: Light Transport Automation of Single Image Deweathering. CVPR 2023
- P16 Singh, A., Ba, Y., Sarker, A., Zhang, H., Kadambi, A., Soatto, S., Srivastava, M. and Wong, A. Depth Estimation from Camera Image and mmWave Radar Point Cloud. CVPR 2023
- P17 Peters, H., Ba, Y. and Kadambi, A. pCON: Polarimetric Coordinate Networks for Neural Scene Representations. CVPR 2023
- P18 Kadambi, A., de Melo, C., Hsieh, C., Srivastava, M. and Soatto, S. Incorporating physics into data-driven computer vision. Nature Machine Intelligence 2023.
<https://doi.org/10.1038/s42256-023-00662-0>

- P19 Chari, P., Ba, Y., Zhou, S., Talegaonkar, C., Athreya, S. and Kadambi, A. On Learning Mechanical Laws of Motion From Video Using Neural Networks. IEEE Access 11 (2023): 30129-30145
- P20 Chari, P., Ba, Y., Athreya, S. and Kadambi, A. MIME: Minority Inclusion for Majority Group Enhancement of AI Performance. ECCV 2022
- P21 Ba, Y., Zhang, H., Yang, E., Suzuki, A., Pfahnl, A., Chandrappa, C., De Melo, C., You, S., Soatto, S., Wong, A. and Kadambi, A. Not Just Streaks: Towards Ground Truth for Single Image Deraining. ECCV 2022
- P22 Ba, Y., Wang, Z., Karinca, D., Bozkurt, O. and Kadambi, A. Style Transfer with Bio-realistic Appearance Manipulation for Skin-tone Inclusive Plethysmography. ICCP 2022
- P23 Vilesov, A., Chari, P., Armouti, A.*, Harish, A. B., Kulkarni, K., Deoghare, A., Jalilian, L. and Kadambi, A. Blending Camera and 77 GHz Radar Sensing for Equitable, Robust Plethysmography. SIGGRAPH 2022 (journal)
- P24 Wang, Z., Ba, Y., Chari, P., Bozkurt, O., Brown, G., Patwa, P., Vaddi, N., Jalilian, L. and Kadambi, A. Synthetic Generation of Face Videos with Plethysmograph Physiology. CVPR 2022
- P25 Pei, S., Chari, P., Wang, X., Yang, X., Kadambi, A. and Zhang, Y. ForceSight: Non-Contact Force Sensing with Laser Speckle Imaging. UIST 2022
- P26 Jalali, B., Zhou, Y., Kadambi, A. and Roychowdhury, V. Physics-AI Symbiosis. ML Science and Technology 2022
- P27 Kadambi, A. Achieving Fairness in Medical Devices. Science 2021 no. 372.6537
- P28 Kalra, A., Brown, B., Stoppi, G., Agrawal, R. and Kadambi, A. Towards Rotation Invariance in Object Detection. ICCV 2021
- P29 Kadambi, A. and Madni, A. Artificial Intelligence: From Ancient Greeks to Self-Driving Cars and Beyond. National Academy of Engineering Bridge 2021
- P30 Ba, Y., Gilbert, A., Wang, F., Yang, J., Chen, R., Wang, Y., Shi, B. and Kadambi, A. Deep Shape from Polarization. ECCV 2020
- P31 Tanaka, K., Mukaigawa, Y. and Kadambi, A. Polarized Non-line-of-sight Imaging. CVPR 2020

- P32 Kalra, A., Taamazyan, V., Rao, S., Venkataraman, K., Raskar, R. and Kadambi, A. Deep Polarization Cues for Transparent Object Segmentation. CVPR 2020 (Oral)
- P33 Tanaka, K., Ikeya, N., Takatani, T., Kubo, H., Funatomi, T., Ravi, V., Kadambi, A. and Mukaigawa, Y. Time-resolved Far Infrared Light Transport Decomposition for Thermal Photometric Stereo. IEEE TPAMI 2020
- P34 Maeda, T., Wang, Y., Raskar, R. and Kadambi, A. Thermal Non-line-of-sight Imaging. IEEE ICCP 2019
- P35 Maeda, T., Kadambi, A., Schechner, Y. and Raskar, R. Dynamic heterodyne interferometry. IEEE ICCP 2018 (Best Paper Award)
- P36 Kadambi, A. and Raskar, R. Rethinking Machine Vision Time of Flight with GHz Heterodyning. IEEE Access 2017
- P37 Kadambi, A., Taamazyan, V., Shi, B. and Raskar, R. Depth sensing using geometrically constrained polarization normals. IJCV 2017 (Best Papers Issue)
- P38 Kadambi, A., Schiel, J. and Raskar, R. Macroscopic Interferometry: Rethinking Depth Estimation with Frequency-Domain Time of Flight. IEEE CVPR 2016 (Oral, 3% acceptance rate)
- P39 Kadambi, A., Zhao, H., Shi, B. and Raskar, R. Occluded Imaging with Time of Flight Sensors. ACM TOG (SIGGRAPH 2016)
- P40 Kadambi, A., Taamazyan, V., Shi, B. and Raskar, R. Polarized 3D: enhanced 3D sensing fusing depth and polarization cues. ICCV 2015 (Oral)
- P41 Naik, N., Kadambi, A., Rhemann, C., Izadi, S., Raskar, R. and Kang, S. A light transport model for mitigating multipath interference in ToF sensors. CVPR 2015
- P42 Bhandari, A., Kadambi, A., Whyte, R., Barsi, C., Feigin, M., Dorrington, A. and Raskar, R. Resolving multi-path interference in time-of-flight imaging via modulation frequency diversity and sparse regularization. Optics Letters 2014
- P43 Kadambi, A., Bhandari, A., Whyte, R., Dorrington, A. and Raskar, R. Demultiplexing Illumination via low-cost sensing and nanosecond coding. ICCP 2014

- P44 Kadambi, A., Whyte, R., Bhandari, A., Streeter, L., Barsi, C., Dorrington, A. and Raskar, R. Coded time of flight cameras: sparse deconvolution to address multipath interference and recover time profiles. ACM TOG (SIGGRAPH Asia 2013)

US Patents Granted (40+ issued to date)

- US1 Systems and methods for transparent object ... US Patent #12,380,568
- US2 Systems and Methods for Generating and Using Visual ... US Patent #12,340,538
- US3 Systems and methods for training pose estimators in ... US Patent #12,293,535
- US4 Sensor fusion between radar and optically polarized ... US Patent #12,249,101
- US5 Systems and methods for picking objects using 3-D ... US Patent #12,172,310
- US6 Systems and methods for a vision guided end effector. US Patent #12,175,741
- US7 Polarization event cameras. US Patent #12,142,025
- US8 Systems and methods for surface normals sensing ... US Patent #12,099,148
- US9 Multi-modal and multi-spectral stereo camera arrays. US Patent #12,069,227
- US10 Systems and methods for using computer vision to ... US Patent #12,067,746
- US11 Systems and methods for high dynamic range ... US Patent #12,020,455
- US12 Systems and methods for pose detection and ... US Patent #12,008,796
- US13 In-hand pose refinement for pick and place automation. US Patent #11,986,955
- US14 Systems and methods for augmentation of sensor ... US Patent #11,982,775
- US15 Multi-aperture polarization optical systems using ... US Patent #11,953,700
- US16 Systems and methods for six-degree of freedom ... US Patent #11,954,886
- US17 Systems and methods for transparent object ... US Patent #11,842,495
- US18 Systems and methods for synthesizing data for training ... US Patent #11,797,863
- US19 Systems and methods for surface modeling ... US Patent #11,699,273
- US20 Systems and methods for high dynamic range imaging ... US Patent #11,689,813
- US21 Systems and methods for camera exposure control. US Patent #11,683,594
- US22 Systems and methods for augmentation of sensor ... US Patent #11,525,906
- US23 Systems and methods for transparent object ... US Patent #11,302,012
- US24 Systems and methods for pose detection and ... US Patent #11,295,475
- US25 Systems and methods for camera exposure control. US Patent #11,290,658
- US26 Systems and methods for surface modeling using ... US Patent #11,270,110
- US27 Systems and methods for characterizing object pose US Patent #11,195,303
- US28 Methods and apparatus for gigahertz time-of-flight imaging. US Patent #11,181,623
- US29 Depth maps with polarization cues. US Patent #10,557,705
- US30 X-ray imaging from temporal measurements. US Patent #10,527,562
- US31 Time-of-flight sensor. US Patent #10,488,520
- US32 Fluorescent lifetime with periodically modulated light. US Patent #10,337,993
- US33 Depth maps with polarization cues. US Patent #10,260,866
- US34 Methods and apparatus for time-of-flight imaging. US Patent #10,191,154

US35 Fluorescence lifetime imaging with pulsed light. US Patent #10,190,983
US36 Methods and apparatus for virtual sensor array. US Patent #9,897,699
US37 Intensity-based depth sensing system and method. US Patent #9,897,698
US38 Methods and apparatus for coded time-of-flight camera. US Patent #9,778,363
US39 Depth sensing using optical pulses and fixed coded ... US Patent #9,638,801
US40 Methods and apparatus for demultiplexing illumination. US Patent #9,451,141
US41 Methods and apparatus for multi-frequency camera. US Patent #9,405,008